Felicity tech tree: light droid

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Abstract

Calculations behind the capabilities of the light droid included in the felicity tech tree.

Contents

1 Motors

The motors identified as ATO-D2BLD100, ATO-D5BLD250 and ATO-D2BLD30-24 are examples motors commercially available from ATO in the 21st century. They are used as a reference to achieve realistic assumptions about the motors the droid may have been built in.

ATO-D5BLD250 motor relevant specifications	
Rated power , Working efficiency	250W , 85%
Holding torque , peak torque	0.8Nm , 2.39Nm
Rated speed	3000 RPM (= 50tr/s)
Weight	3kg
Physical dimensions (body only)	70mm × 86mm × 86mm
Physical dimensions (with rotor)	107mm × 86mm × 86mm

ATO-D2BLD100 motor relevant specifications	
Rated power , Working efficiency	100W , 85%
Holding torque , peak torque	0.32Nm , 0.96Nm
Rated speed	3000 RPM (= 50tr/s)
Weight	2.5kg
Physical dimensions (body only)	95mm × 57mm × 57mm
Physical dimensions (with rotor)	120mm × 57mm × 57mm

ATO-D2BLD30-24 motor relevant specifications	
Rated power , Working efficiency	30W , 85%
Holding torque , peak torque	0.1Nm , 0.29Nm
Rated speed	3000 RPM (= 50tr/s)
Weight	1.5kg
Physical dimensions (body only)	60mm × 60mm × 60mm
Physical dimensions (with rotor)	85mm × 60mm × 60mm

25:1 planetary gearing can be used to reduce the speed from 50 down to 2 revolutions per second, a reasonable speed for the movement of the droid's limbs, at the same time correspondingly increasing the holding torque and peak torque by 25 times.

ATO-D2BLD30-24 motor appears to be the best fit given the present space constraints, but is too weak. ATO-D5BLD250 will require additional space, but is most likely to allow the droid to lift itself without requiring crippling gear ratios.

If 2 motors (one for bending, one for rotating the limb) are to be present in a limb segment, then the segment may have dimensions of at least 86mm × 86mm × 214mm, in practice more likely 250mm in length. A differential gearbox can be used for the gearing, such that when both motors turn in the same direction they bend the joint, and when they turn in opposite directions they rotate the limb. That way the combined strength of both motors can be used for each movement instead of one of the motors taking the full load and the other remaining unused. Given the 250mm length and 25:1 gear ratio, the two motor's combined 1.6Nm holding torque could exert 160N of force, and the combined peak torque of 4.78Nm could exert 478N of force.

Given that there are 4 limbs for locomotion use, this **gives the droid a lifting force of at least 640N consistently or 1912N in short bursts**, equivalent to the weight of 65.26kg and 194.97kg in a 1 gee gravity environment. This is doubled if the remaining 4 limbs are also used for locomotion rather than manipulation. In practice the droid's actual carrying capacity would likely exceed these forces severalfold, since the motors would usually be only dealing with the small fractions of the forces that are actually bending the limb instead of being transferred through its strongest angles.

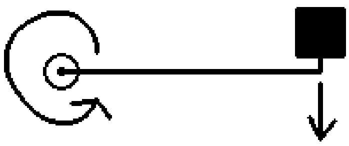
If on average there is 1 motors in each limb segment (alternating between 2 and 0), 5 segments in each limb and 8 limbs in total, plus 4 segments making up the spine, there would be **44 motors, adding up to 132kg** and a theoretical maximum 33kW power consumption.

The details need not stay this specific for the actual design, but this would seem like a good estimate.

$$\frac{2 \cdot 0.8Nm \cdot 25}{0.25m} = 160N \qquad \frac{2 \cdot 2.39Nm \cdot 25}{0.25m} = 478N \qquad (5 \cdot 8 + 4) \cdot 3kg = \mathbf{44} \cdot 3kg = \mathbf{132kg}$$

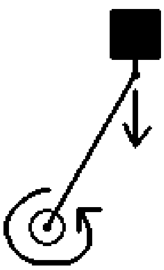
$$\frac{4 \cdot 160N}{9.807m/s^2} = \frac{\mathbf{640N}}{9.807m/s^2} = 65.2595kg \qquad \frac{4 \cdot 478N}{9.807m/s^2} = \frac{\mathbf{1912N}}{9.807m/s^2} = 194.9628kg$$

2 Real carrying capacity



When calculating the carrying capacity in the previous section, I used the situation showed on the right. Imagine that the motor is applying a torque of 10Nm, the pole is 2m long, and the mass on the end has a weight of 5N. In that setup, the weight is perfectly balanced against the torque of the motor and the length of the pole.

Bringing in the real numbers, given the gearing and motors, each limb segment is given a torque of 80Nm. The segments are also said to be 250mm long. With these numbers applied to the setup shown on the left, the weight at the end could at maximum be 160N.



A more realistic scenario would look like this. A simple exercise to feel just how much of a difference it makes would be to stand normally for a while, and then try standing with your knees bent at a right angle.

In the image on the left, the pole is 30 degrees off of vertical. If it were fully vertical, theoretically the motor would not need any torque at all to support the weight, but reality often supplies disturbances that have to be taken care of.

Given my earlier example of a 10Nm motor with a 2m long pole, at this angle it would be able to support a 10N weight, that is twice as much.

The relationship between maximum weight and angle can be described like this:

$$W_{max} = \frac{T}{L \cdot sin(\theta)}$$

where:

- W_{max} is the biggest weight that can be put on the end of the pole [N]
- T is the torque of the motor [Nm]
- L is the length of the pole [m]
- θ is the angle of the pole compared to being fully vertical

So I said the droid is capable of carrying 640N consistently, and lifting up to 1912N in short bursts. That would actually be what I'll call a guaranteed lifting/carrying capacity, a weight that it can lift regardless of the angle of its limbs. Assuming that regular walking doesn't require leg angles beyond 30 degrees away from vertical, the real carrying capacity is double that of what I mentioned.

guaranteed carrying capacity (consistent)	640N	65.26kgf
guaranteed lifting capacity (momentary)	1912N	194.97kgf
nominal carrying capacity (consistent)	1280N	130.52kgf
nominal lifting capacity (momentary)	3824N	389.94kgf

single limb: $\frac{40Nm}{0.25m \cdot sin(30^o)} = \frac{40Nm}{0.25m \cdot 0.5} = 320N$
all 4: $\frac{640N}{sin(30^o)} = \frac{640N}{0.5} = 1280N$

3 Batteries

Solid state thin film lithium ion todo from existing documentation not yet transcribed onto the wiki.

The droid contains a total of 20 battery modules, each 1MJ and 0.5kg for a total of **20MJ** and **10kg**.

4 Stat card